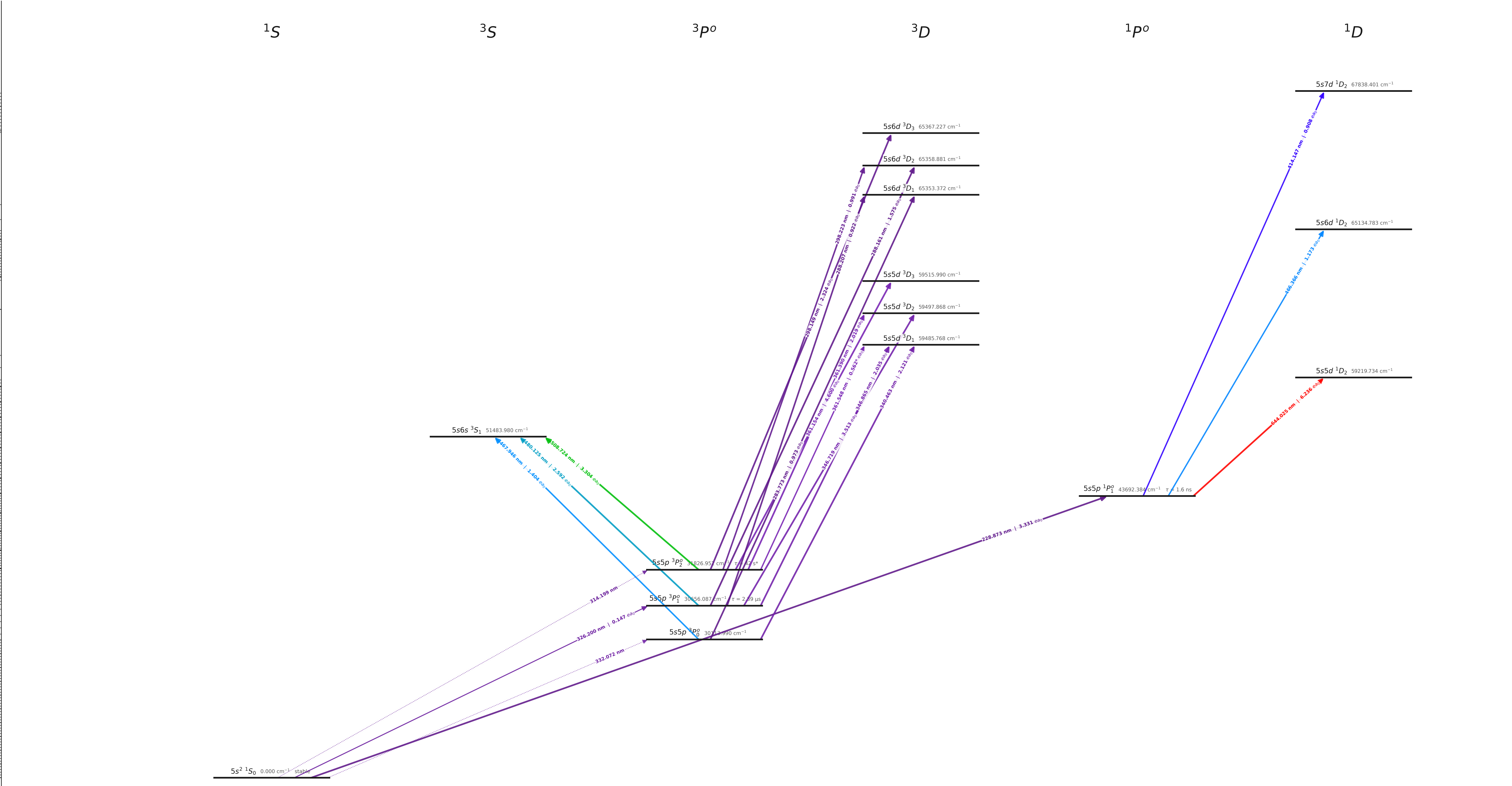


# Cadmium

Cd (Cd I). Strongest classified transitions below 2 μm (NIST): 15 levels, 21 lines.

Cd<sup>+</sup> ionisation limit 72540.05 cm<sup>-1</sup> = 8.99382 eV (137.855 nm)



## Key transitions

| Transition                                                          | λ vac (nm) | λ air (nm) | ν (THz)         | [U  er  U']  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|---------------------------------------------------------------------|------------|------------|-----------------|---------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 1P <sub>1</sub> <sup>o</sup> | 228.8729   | 228.8023   | 1309.864341(20) | 3.331                           | 6.250e+08            | 99.47          | 100        | 228.9 nm resonance line: first-stage MOT / slowing / imag... |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 314.1991   | 314.1081   | 954.1480        | -                               | -                    | 3.789e-09      | -          | 314 nm ultra-narrow                                          |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>1</sub> <sup>o</sup> | 326.1995   | 326.1055   | 919.046234(21)  | 0.147                           | 4.184e+05            | 0.06659        | 100        | 326.1 nm intercombination line: narrow-line MOT              |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>0</sub> <sup>o</sup> | 332.0716   | 331.9761   | 902.79462(6)    | -                               | -                    | -              | -          | 332 nm optical lattice clock transition                      |
| 5s5p 3P <sub>2</sub> <sup>o</sup> - 5s5d 3D <sub>3</sub>            | 361.1537   | 361.0507   | 830.096573(15)  | 4.600                           | 1.300e+08            | -              | -          | 361.2 nm metastable MOT                                      |
| 5s5p 3P <sub>2</sub> <sup>o</sup> - 5s5d 3D <sub>1</sub>            | 361.5484   | 361.4453   | 829.1904        | 0.562*                          | 4.510e+06            | -              | -          | 361.5 nm                                                     |
| 5s5p 3P <sub>1</sub> <sup>o</sup> - 5s6s 3S <sub>1</sub>            | 480.1254   | 479.9912   | 624.4045        | 2.592                           | 4.100e+07            | -              | -          | 480.0 nm optical pumping / repumping                         |

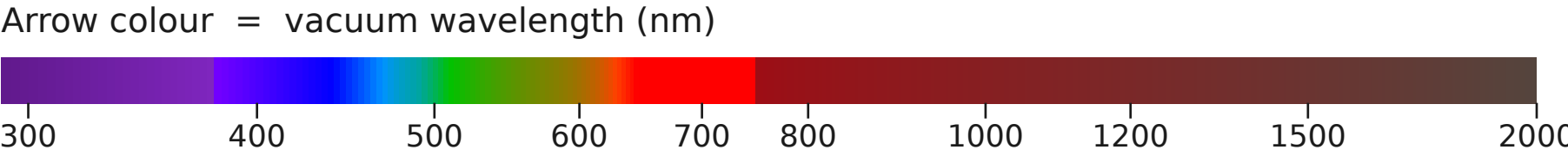
λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |[U||er||U']| in ea<sub>0</sub>

Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## <sup>111</sup>Cd hyperfine structure (I = 1/2)

| Level                             | A (MHz)       | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|-----------------------------------|---------------|---------|--------------------------------------------------------|
| 5s5p 3P <sub>0</sub> <sup>o</sup> | -4123.813(13) | -       | F=1/2: +4123.81 F=3/2: -2061.91                        |
| 5s5p 3P <sub>2</sub> <sup>o</sup> | -3292.9364(8) | -       | F=3/2: +4939.40 F=5/2: -3292.94                        |
| 5s5p 1P <sub>1</sub> <sup>o</sup> | 188.5(22)     | -       | F=1/2: -188.50 F=3/2: +94.25                           |
| 5s6s 3S <sub>1</sub>              | -7750.8(9)    | -       | F=1/2: +7750.80 F=3/2: -3875.40                        |
| 5s5d 3D <sub>3</sub>              | -2386.297(26) | -       | F=5/2: +4772.59 F=7/2: -3579.45                        |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                          | λ vac (nm) | Isotopes | Shift (MHz) |
|---------------------------------------------------------------------|------------|----------|-------------|
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 1P <sub>1</sub> <sup>o</sup> | 228.873    | 106-114  | 1818(4)     |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 1P <sub>1</sub> <sup>o</sup> | 228.873    | 108-114  | 1336(3)     |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 1P <sub>1</sub> <sup>o</sup> | 228.873    | 110-114  | 865(3)      |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 1P <sub>1</sub> <sup>o</sup> | 228.873    | 112-114  | 408(3)      |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 1P <sub>1</sub> <sup>o</sup> | 228.873    | 116-114  | -316(3)     |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 1P <sub>1</sub> <sup>o</sup> | 228.873    | 111-114  | 805.0(3)    |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 1P <sub>1</sub> <sup>o</sup> | 228.873    | 113-114  | 344.9(5)    |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 314.199    | 106-114  | 1916(4)     |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 314.199    | 108-114  | 1404(3)     |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 314.199    | 110-114  | 913.7(21)   |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 314.199    | 112-114  | 429.7(16)   |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 314.199    | 116-114  | -320.2(17)  |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 326.200    | 106-114  | 1913.0(10)  |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 326.200    | 108-114  | 1402.4(10)  |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 326.200    | 110-114  | 914.7(10)   |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 326.200    | 112-114  | 429.9(10)   |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>2</sub> <sup>o</sup> | 326.200    | 116-114  | -321.5(10)  |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>1</sub> <sup>o</sup> | 326.200    | 111-114  | 861.6(10)   |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>1</sub> <sup>o</sup> | 326.200    | 113-114  | 370.8(10)   |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>0</sub> <sup>o</sup> | 332.072    | 106-114  | 1911(5)     |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>0</sub> <sup>o</sup> | 332.072    | 108-114  | 1404(4)     |
| 5s <sup>2</sup> 1S <sub>0</sub> - 5s5p 3P <sub>0</sub> <sup>o</sup> | 332.072    | 110-114  | 914.1(21)   |

Shift = ν(first isotope) - ν(second isotope).